same initial point, we can move one of the vectors suitably and make them to have same initial point. This leads us to the following Definition 8.11.

Let $\vec{a}$ and $\vec{b}$ be two vectors with the same initial point O. Let A and B be the terminal points of $\vec{a}$ and $\vec{b}$ respectively.

Complete the parallelogram $OACB$. Then the vector $\overrightarrow{OC}$ is defined as the sum of the vectors $\vec{a}$ and $\vec{b}$. Thus

Definition 8.11 (**Parallelogram law of addition**)
In a parallelogram $OABC$ if $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represents two adjacent sides, then the diagonal $\overrightarrow{OC}$ represents their sum (see Fig. 8.17).

Though we have two definitions for addition of vectors, they are one and the same. Definition 8.10 is defined using the triangle law for addition of vectors and Definition 8.11 is defined using the parallelogram law for addition of vectors:

In a triangle ABC if $\overrightarrow{AB}$ and $\overrightarrow{BC}$ represent two sides, then the third side $\overrightarrow{AC}$ represents their sum.

8.4.2 Difference between two Vectors

Now let us see how to subtract one vector from another vector.

Definition 8.12
Let $\vec{a}$ be a vector. Then the **reverse** of $\vec{a}$, denoted by $-\vec{a}$, is defined as the vector having the magnitude of $\vec{a}$ and the direction opposite to the direction of $\vec{a}$.

Notice that if $\overrightarrow{AB} = \vec{a}$, then $\overrightarrow{BA} = -\vec{a}$.

Geometrical interpretation of difference between two vectors

Let $\vec{a}$ be a vector with initial point P and terminal point Q. Let $\vec{b}$ be the vector with initial point Q and terminal point P. The magnitude of both of the vectors is the length of the line segment joining P and Q. So they have the same magnitude. But clearly they have opposite directions. So $\vec{b}$ is equal to $-\vec{a}$.

If $\vec{a}$ and $\vec{b}$ are two vectors, then the vector $\vec{a} - \vec{b}$ is defined as the sum of the vectors $\vec{a}$ and $-\vec{b}$. That is $\vec{a} + (-\vec{b})$.

We can view the vector $\vec{a} - \vec{b}$ geometrically. Let $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent the vectors $\vec{a}$ and $\vec{b}$ respectively (Fig. 8.18) . Draw AC parallel to OB with $AC = OB$. Then $\overrightarrow{AC}$ is equal to $\vec{b}$. Extend the line CA to D so that $CA = AD$. Then $\overrightarrow{AD}$ is equal to $-\vec{b}$. Thus $\vec{a} + (-\vec{b}) = \overrightarrow{OD}$. Hence $\vec{a} - \vec{b} = \overrightarrow{OD}$ (Fig. 8.19).

Let us complete the parallelogram $OACB$. We observe that BA and OD are parallel and they have equal length. Thus the two vectors $\overrightarrow{BA}$ and $\overrightarrow{OD}$ are equal. So we may consider $\overrightarrow{BA}$ as $\vec{a} - \vec{b}$. This shows that if the sides $\overrightarrow{OA}$ and $\overrightarrow{OB}$ of the parallelogram $OACB$ represent the vectors $\vec{a}$ and $\vec{b}$ respectively, then the diagonal $\overrightarrow{BA}$ will represent the vector $\vec{a} - \vec{b}$. (Fig. 8.20). We note that we have already seen that the diagonal $\overrightarrow{OC}$ represents the vector $\vec{a} + \vec{b}$.

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